

2. Docker

- Create Docker file(s)
- Run app using Docker containers
- Expose required ports
- Ensure containers auto-start on reboot

```
ubuntu@ip-172-31-72-202:~$ mkdir docker
ubuntu@ip-172-31-72-202:~$ cd docker
ubuntu@ip-172-31-72-202:~/docker$ vi app.py
ubuntu@ip-172-31-72-202:~/docker$ vi requirements.txt
ubuntu@ip-172-31-72-202:~/docker$ vi dockerfile
```

Vi app.py

```
root@ip-172-31-72-202: /home/ubuntu/docker
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Dockerized Python App is Running Successfully!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Vi requirements.txt

```
root@ip-172-31-72-202: /home/ubuntu/docker
flask

~
~
~
~
~
~
~
~
```

Vi dockerfile

```

root@ip-172-31-72-202: /home/ubuntu/docker
# Use official Python image
FROM python:3.11-slim

# Set working directory
WORKDIR /app

# Copy files
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY app.py .

# Expose application port
EXPOSE 5000

# Run the app
CMD ["python", "app.py"]

~
~

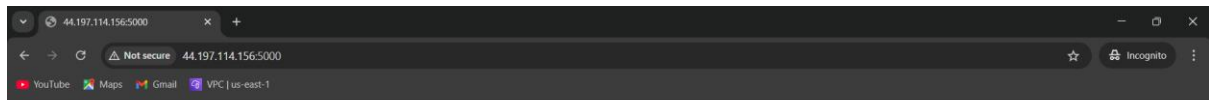
```

```

root@ip-172-31-72-202:/home/ubuntu/docker# docker build -t python-docker-app .
[+] Building 9.8s (10/10) FINISHED                                docker:default
=> [internal] load build definition from dockerfile                0.0s
=> => transferring dockerfile: 311B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 0.4s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                       0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:db27ce7778e5f581d5d97 3.3s
=> => resolve docker.io/library/python:3.11-slim@sha256:db27ce7778e5f581d5d97 0.0s
=> => sha256:d85099f0969e8b2306770a12dffcd300208cc3a18b876c0ad3d 250B / 250B 0.0s
=> => sha256:64faa99400e1388ed9f202917ada9fac34fd46c950d40cd4 1.29MB / 1.29MB 0.1s
=> => sha256:8cbc47ff628d718fb76f7fca9897e4e8b607a4f543008c 14.36MB / 14.36MB 0.3s
=> => sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc 29.78MB / 29.78MB 0.8s
=> => extracting sha256:0c8d55a45c0dc58de60579b9cc5b708de9e7957f4591fc7de941b 1.1s
=> => extracting sha256:64faa99400e1388ed9f202917ada9fac34fd46c950d40cd410236 0.2s
=> => extracting sha256:8cbc47ff628d718fb76f7fca9897e4e8b607a4f543008cdee7607 0.8s
=> => extracting sha256:d85099f0969e8b2306770a12dffcd300208cc3a18b876c0ad3dc 0.2s
=> [internal] load build context                                  0.0s
=> => transferring context: 291B                                      0.0s
=> [2/5] WORKDIR /app                                             0.4s
=> [3/5] COPY requirements.txt .                                  0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt      4.2s
=> [5/5] COPY app.py .                                           0.1s
=> exporting to image                                             1.2s
=> => exporting layers                                              0.8s
=> => exporting manifest sha256:30fc367e126ad79b2c6c171c58f7fb5fc16dbc3c442a0 0.0s
=> => exporting config sha256:d3150fcbe714240d6e48f0aa6bcbdd7e311bcadf462abe8f 0.0s
=> => exporting attestation manifest sha256:b486d3c3bb069238587e8229f10d9d691 0.0s
=> => exporting manifest list sha256:ccf331dc0031946d7f1c0ccc919cb6d77b633a17 0.0s
=> => naming to docker.io/library/python-docker-app:latest      0.0s
=> => unpacking to docker.io/library/python-docker-app:latest   0.3s
root@ip-172-31-72-202:/home/ubuntu/docker#

```

```
root@ip-172-31-72-202:/home/ubuntu/docker# docker run -d -p 5000:5000 --name my-python-app python-docker-app
70a9e6168fd5043a2d69955684b4cb1a2970073324eaed260e3de7b0eddd41a5
root@ip-172-31-72-202:/home/ubuntu/docker#
```



```
root@ip-172-31-72-202:/home/ubuntu/docker# docker run -d \
-p 5000:5000 \
--name my-python-app \
--restart unless-stopped \
python-docker-app
docker: Error response from daemon: Conflict. The container name "/my-python-app" is already in use by container "70a9e6168fd5043a2d69955684b4cb1a2970073324eaed260e3de7b0eddd41a5". You have to remove (or rename) that container to be able to reuse that name.
```

```
Run 'docker run --help' for more information
root@ip-172-31-72-202:/home/ubuntu/docker# docker update --restart unless-stopped my-python-app
my-python-app
root@ip-172-31-72-202:/home/ubuntu/docker# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
70a9e6168fd5   python-docker-app "python app.py"         4 minutes ago Up 4 minutes  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp
root@ip-172-31-72-202:/home/ubuntu/docker# docker stop my-python-app
my-python-app
root@ip-172-31-72-202:/home/ubuntu/docker#
root@ip-172-31-72-202:/home/ubuntu/docker#
```

```
root@ip-172-31-72-202:/home/ubuntu/docker# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
root@ip-172-31-72-202:/home/ubuntu/docker#
```